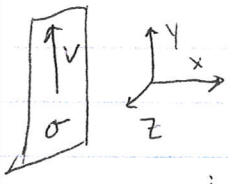
Problem: infinite sheet of charge starts moving upward with speed v at $t=0$. (motionless before) find $\vec{B}(\vec{r})$, and $\vec{E}(\vec{r})$.



We start from $\vec{A}(\vec{r}) = \int d^3\vec{r}\,' \dfrac{\vec{j}(t', \vec{r}\,')}{|\vec{r}-\vec{r}\,'|} \dfrac{\mu_0}{4\pi}$

where t' is $t' = t - \dfrac{|\vec{r}-\vec{r}\,'|}{c}$

$\vec{r}\,'$ is position of source charges

$\vec{r}$ is observation point

$\vec{j} = \rho\vec{v}$ in general so for moving sheet of charge in yz plane

$\vec{j} = \sigma v\, \delta(x)\, \Theta(t)\, \hat{y}$

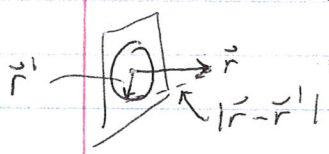
$\Theta(t)$ is step function $\vec{j} = 0$ for $t < 0$

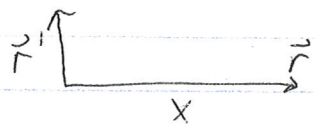
$t = t' + \dfrac{|\vec{r}-\vec{r}\,'|}{c}$

Now we have

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{-\infty}^{\infty} dy\,dz\, \frac{\sigma v\, \Theta\left(t + \frac{|\vec{r}-\vec{r}\,'|}{c}\right)}{|\vec{r}-\vec{r}\,'|}\, \hat{y}$$

Since the fields created at $t=0$ take time to propagate an observer at $\vec{r}$ only sees the field from an expanding circle on the sheet. By symmetry we can limit $\vec{r}$ to points on the x-axis since field is a plane wave expanding outward





The maximum radius of the observed area is $r'_{max} = \sqrt{c^2t^2 - x^2}$ because $|\vec{r}-\vec{r}\,'| = \sqrt{x^2 + r'^2}$

We change $\int dy\,dz$ to integrating circular area $\int 2\pi r\,dr$ and the step function creates the limits $0 \to r'_{max}$

$$\vec{A}(x) = \frac{\mu_0 \sigma v\, \hat{y}}{2} \int_0^{\sqrt{c^2t^2 - x^2}} \frac{r'\,dr'}{\sqrt{x^2 + r'^2}}$$

Substitute $u = x^2 + r'^2$ $\quad du = 2r'dr'$

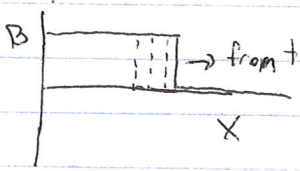
$$\vec{A}(x) = \frac{\mu_0 \sigma v}{4} \hat{y} \int_{x^2}^{c^2t^2} \frac{du}{\sqrt{u}}$$

$$\vec{A} = \frac{1}{2} \mu_0 \sigma v (ct - x) \hat{y}$$ This of course is only valid for $t > x/c$

Now we can find $\vec{E}$ and $\vec{B}$

$$\vec{B} = \nabla \times \vec{A} = \frac{\partial}{\partial x} A_y \hat{z} = \boxed{-\tfrac{1}{2} \mu_0 \sigma v \hat{z}}$$ for $x < ct$

so $\vec{B}$ is constant within expanding "shock wave"



→ front is moving with speed c

This is the same result you get from Amperel's Law and right-hand rule for sign

answer to original problem

extra discussion from class

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \tfrac{1}{2} \mu_0 \sigma v c \hat{y} = \boxed{\frac{\sigma v}{2\epsilon_0 c} \hat{y}}$$

This $\vec{E}$ field arises purely from the "boost" at $t = 0$

$\vec{E}$ would be 0 if sheet had constant current

There is also the electrostatic field $\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$ if sheet has net charge.

Poynting vector $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

$$\vec{S} = \frac{1}{4} \mu_0 c \sigma^2 v^2 \hat{x}$$ energy from boost flows away from sheet

including electrostatic fields

$$\mu_0 \vec{S} = \left[\frac{\sigma}{2\epsilon_0} \hat{x} \times -\frac{\mu_0 \sigma v}{2} \hat{z} \right]$$

$$\vec{S} = \frac{\sigma^2 v}{4\epsilon_0} \hat{y}$$ electrostatic fields create Poynting vector that follows charge velocity